

AI Engineer

Step 3: Learn Deep Learning

CareerCompass • Detailed Concept Notes • Research-backed learning guide

Concept & Learning Objective

Deep learning uses neural networks with learned representations. You should understand the training loop before jumping to large pretrained models: inputs → forward pass → loss → gradients → optimizer update → repeat.

Study dense networks first, then CNNs for images and transformer/attention concepts for sequence and multimodal workloads. Always compare training behavior with validation behavior to detect overfitting.

Core Concepts — Detailed Breakdown

- Neuron/layer: weighted inputs plus bias transformed by an activation.
- Activations: ReLU and related functions; understand why nonlinearities are necessary.
- Forward pass: compute predictions.
- Loss: quantify prediction error.
- Backpropagation: efficiently compute gradients through the computational graph.
- Optimizers: SGD, momentum and Adam-style adaptive methods.
- Batch/epoch/learning rate.
- Regularization: dropout, weight decay, augmentation and early stopping.
- CNN concepts: convolution, kernels, feature maps and pooling.
- Attention/transformers: query/key/value intuition and why attention supports contextual relationships.

What You Should Be Able To Do

- Train a small neural network.
- Plot training vs validation loss.
- Change learning rate and observe convergence.
- Identify overfitting.
- Explain the difference between parameters and hyperparameters.

Hands-on Project / Practice

- Train a small image classifier. Run two controlled experiments changing one hyperparameter at a time and document the effect.

Recommended References

- TensorFlow Learn — <https://www.tensorflow.org/learn>
- PyTorch Tutorials — <https://pytorch.org/tutorials/>
- Deep Learning Course for Beginners — <https://www.youtube.com/watch?v=HJd1I3FdSnY>
- TensorFlow 2.0 Complete Course — <https://www.youtube.com/watch?v=tPYj3fFJGjk>

Note: These notes are designed to teach the concepts and provide a practical learning path. Always prefer the linked official documentation for current APIs, SDK versions and platform-specific release requirements.